Anomaly Detection Report

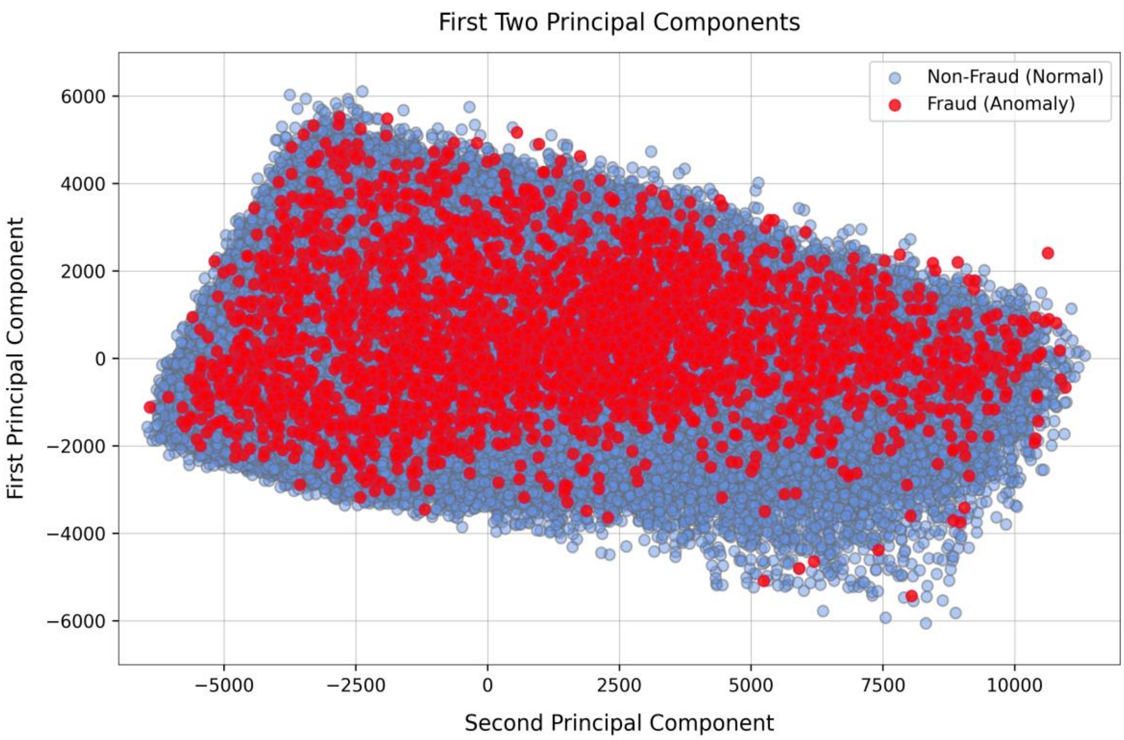
Unsupervised Anomaly Detection Results

Data	True Positive Rate	AUC Score	Balanced Accuracy	F1 Score
Train	0.107	0.589	0.508	0.027
Validation	0.113	0.584	0.507	0.025
Test	**0.105**	0.592	**0.508**	0.028

Note: The dataset was partitioned into to a 60/10/30 train/validate/test split

The results above suggest that the Isolation_Forest algorithm was **unable to classify fraudulent bank account records**. This contradicts our initial hypothesis that the small proportion of fraud records (~1%) could by identified using anomaly detection.

To understand why the Isolation_Forest algorithm performed poorly, we can generate **principal components** from the test data and visualize their class labels in a scatter plot.



The figure above shows that there is **no defined spatial separation or cluster** for records of each class. Instead, all the records are grouped in one big cluster.

This suggests that fraudulent bank accounts have similar feature values to regular accounts. Meaning that a **more sophisticated model is required** to predict fraud records.